ACS WORK INTEGRATED LEARNING | TELECOMMUNICATIONS

Final Report

Customer Churn Analysis for a Telecommunications Company

Document type | Project report

Data analysed | 6,741 telecommunications customer records

Project stage | Completed modelling and business interpretation

PROJECT TEAM

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1. Executive Summary

Exploratory analysis, K-Means Segmentation and predictive modelling was done to understand customer churn in the project. A total of 1,791 customers (out of 6,741 who came from the supplied CSV files) have churned, which is a 26.57% churn rate. Cleaned data file with 12 fields of explanation and outcome. In addition to the original file, there is a companion file containing historical positions of the clusters and PCA coordinates.

There is significant disparity in churn between the different segments published. Cluster 3 has 2,146 customers and 48.84% churn, compared with 4.80% in Cluster 1. These results are now used to update the sample with more current rankings. The values of tenure and charges are already standardised and, therefore, do not signify months or amounts of money in their original form.

Logistic regression and random forest were compared to the ANN. It was able to achieve held-out ROC-AUC 0.810 with PR-AUC 0.587, at the selected threshold, it detected 91.90% of the churners with only 39.98% alert precision. Contact priority should be based on prediction and interventions should be based on Segmentation that should be tested in a well-controlled pilot for economic value.

2. Project Background and Business Problem

Customer attrition has an impact on the recurring revenue, Value of Future Customers and the acquisition costs of new customers. Different situations occur among the different groups of customers for the telecommunications provider. In the case of a long-serving customer base, they will look for reliable support and in the case of a fairly new customer base on an expensive plan, they will be considering other options. These differences would be ignored by a single discount for all and the single discount might lead to a decrease in revenue without influencing behaviour.

These are two inter-related business questions that are being addressed in the project. For each query, you'll find out what is the t-score for the group before you receive the answer.First, you will see the t-score for the group for each query before your answer. Secondly, to what extent can the existing customer information match the individuals who will be at risk on their accounts? Segmentation is helpful to understand overarching profiles. Predictive classification helps to prioritise on an individual level. Not to be viewed as causal evidence. A high-risk cluster is not the same as high-risk prediction score; a high-risk cluster is not a marker of why members are likely to leave, but a prediction score is not necessarily a marker of what intervention is going to prevent members from leaving.

Analysis is based on historical, encoded information and not a real-time customer-management system. Therefore, the product delivered must be a prototype system for decision making and an implementation proposal and can not simply claim to be a functioning retention programme. It is this distinction which leads the interpretation of performance and recommendations throughout this report.

3. Project Objectives and Scope

1. Validate the uploaded customer records, their encoded columns and the relationship between the two datasets.
2. Describe overall churn and compare profiles across four existing customer clusters.
3. Train a new leakage-conscious K-Means representation on the training set only.
4. Build an artificial neural network and compare it with logistic regression and random forest.
5. Evaluate accuracy, precision, recall, F1, ROC-AUC, PR-AUC and confusion-matrix counts on unseen customers.
6. Translate evidence into a practical retention pilot, monitoring plan and stakeholder measures.

The scope includes the prepared project files, Exploratory analysis, Segmentation comparison, Predictive training and Test evaluation & Business Recommendations. Production system integration and direct observation of contacts with customers, as well as realised revenue savings, are not included. The project also

does not reproduce the raw customer charges and tenure values that are not provided, from the standardised fields.

4. Team Members and Roles

The project team was Sanjana (Project Manager), Aadesh (Data Analyst for Clustering), Sateesh (Data Analyst for Predictive Modelling), Uday (Business Analyst) and Mehrab (Data Engineer). They will be assigned to the roles of project co-ordinator, data preparation, customer segmentation and predictive modelling etc. and will share the findings of the model with various stakeholders. These responsibilities relate to the technical work and the customer retention decisions detailed out in this report.

Table 1. Project team members and responsibilities

Team Member	Role	Primary Responsibility
Sanjana	Project Manager	Project coordination and delivery oversight
Aadesh	Data Analyst (Clustering)	K-Means clustering and customer segmentation
Sateesh	Data Analyst (Predictive Modelling)	ANN modelling and evaluation of predictive results
Uday	Business Analyst	Business interpretation and stakeholder recommendations
Mehrab	Data Engineer	Data preparation and validation

This workflow is a separation of the technical delivery from the operational judgement. Before using it, the business stakeholders need to establish contact-costs and missed-churn risks that are acceptable to them: a high recall approach can generate many false-positive alerts.

5. Dataset and Data Preparation

The provided and cleaned dataset has 6741 rows and 13 columns. There are 12 columns which are predictors and one column of Churn. The accompanying grouped file contains the same values and the same rows as the original file along with Cluster, PCA1 and PCA2. During verification no missing data and no exact duplicate records were detected. The target is off balance as there are 4950 retained accounts and 1791 churned accounts. Data manipulation has been documented by Pandas Development Team (n.d.) to verify these counts and data consistency in fields.

Table 2. Validated source data summary

Dataset property	Observed value	Implication
Customer records	6,741	One row per encoded customer observation
Explanatory columns	12	Input fields before new clustering
Actual churn cases	1,791 (26.57%)	Minority outcome
No-churn cases	4,950 (73.43%)	Majority outcome
Existing clusters	Four (0–3)	Descriptive comparison only
Train / validation / test	4,313 / 1,079 / 1,349	Separate development and final assessment

Predictors such as gender, senior, dependents, tenure, phone services, month to month, internet-service indicators and contract indicators are included. Some of them are even one-hot encoded and others like tenure and monthly charges are already converted to standardised numerical values. The upstream scaling parameters are not given. It doesn't facilitate an accurate conversion to months or dollars and there is still a chance that there could have been some transformations that were done prior to the train and test split.

The update pipeline is able to process the extra pre-processing and K-Means segmentation only for the training subset. Published Cluster and PCA fields are still useful to describe the available data set, but are not fed

to the training as predictors. This separation prevents copying information from test observations via the published clustering workflow with the full data.

6. Methodology Overview

Analysis is based on an enacted and reproducible modelling sequence. Initial source audit reports on column names and missingness, and consistency between cleaned and clustered file and label values. Churn Exploration is the occurrence of churn and previously documented customer segments. The stratified splitting then puts 4,313 customers to be part of training, 1,079 customers to be part of validation and 1,349 customers to take part in held-out testing. Final proportions are ~64, 16 and 20.

Table 3. Analytical stages and decision points

Stage	Method	Decision output
Exploration	Label, tenure and segment summaries	Baseline risk profile
Segmentation	Four-cluster K-Means trained on development data	New cluster indicators without test leakage
Classification	ANN, logistic regression and random forest	Comparable churn probabilities
Model selection	Validation average precision	ANN selected before test review
Decision threshold	Validation F2 optimisation	Operating point reflecting recall priority
Business translation	Risk, value and campaign constraints	Pilot strategy with measurable outcomes

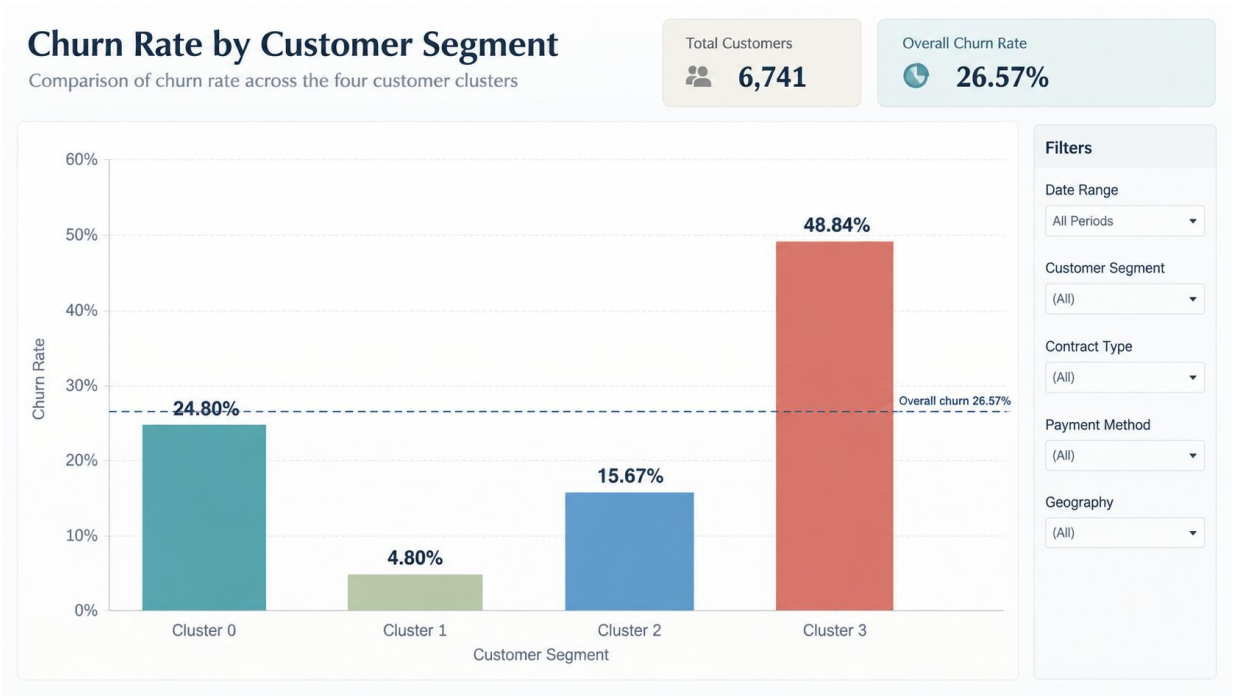
The 12 predictors in the original data set are augmented by four newly fitted cluster indicators to provide the same 16 predictors per supervised model. The top validation averaged precision is used to select the ANN. Optimal operating point is selected separately by maximising validation F2, which puts more emphasis on the recall. These decisions use a held out set which is used only after the time of these decisions. The Scikit-learn Developers (n.d.) share how doing so promotes more meaningful comparison.

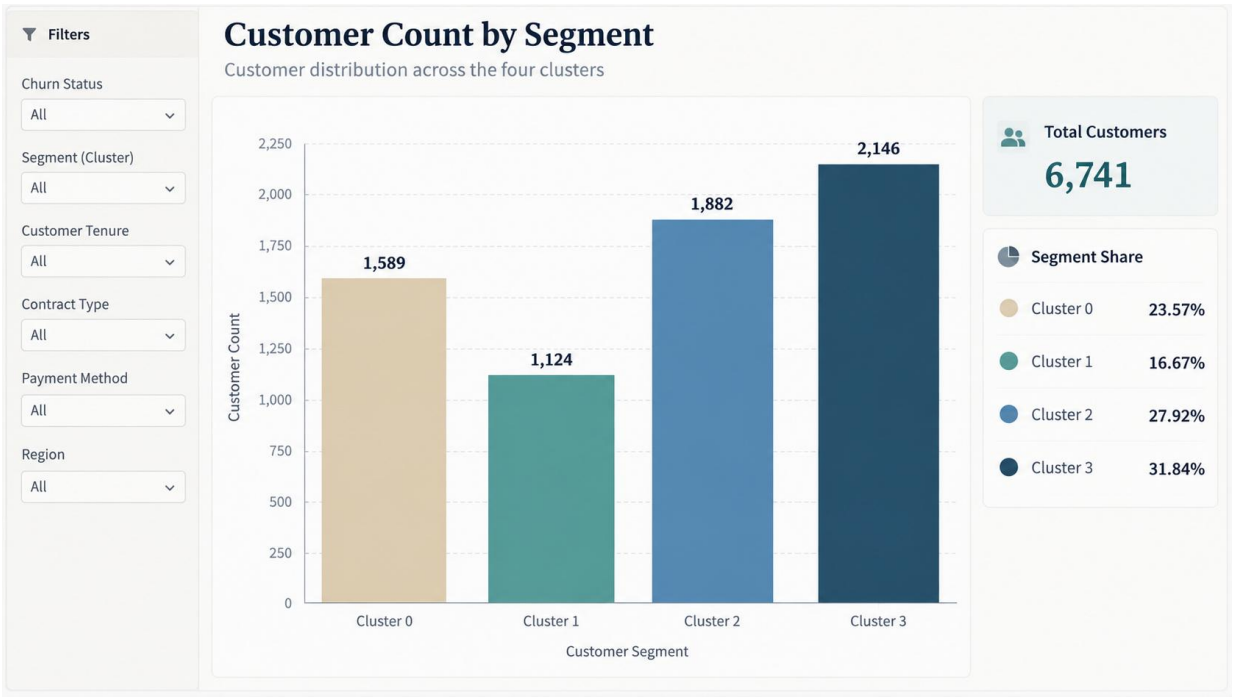
7. K-Means Clustering Analysis

A total of four published customer groups were created using K-Means based on the repository supplied. The labels are informative and are used for the description of the business, they are also verified against the latest csv, however they are not transferable with the new type of clusters fitted in to the predictive pipeline. The difference is because the original groups have been established before the split in the predictive data. Any modelling workflow prohibits the use of full-dataset derived columns and rates can be displayed as descriptive.

Table 4. Published customer segment comparison

Cluster	Customers	Share	Churn rate	Standardised charges	Relative tenure
0	1,589	23.57%	24.80%	−1.028	Shorter
1	1,124	16.67%	4.80%	−0.989	Longer
2	1,882	27.92%	15.67%	+0.947	Longer
3	2,146	31.84%	48.84%	+0.562	Shorter





The commercial contrast is of importance. The Cluster 3 segment has the highest number of customers, and the highest percentage of churn observed. Of the 1,791 churn cases in the dataset, 1,048 cases are contained in it. There are only 54 observed customers that churn in Cluster 1, that accounts for more than one thousand customers. Charges in Cluster 2 are above average at the monthly level but churn in the cluster is less than expected. This is a direct contradiction to the ranking in the older example document, and adds to the importance of a decision being made based on the provided csv and not on the re-used narrative.

7.1 Segment Interpretation

Cluster 0: Lower-Charge, Shorter-Tenure Customers

There are 1,589 customers in cluster 0 with a churn rate of 24.80% which is slightly less than the overall 26.57%. It has standardised charges and tenure which are comparatively low. This communication trial may be feasible and is likely to be a trial based on affordability but affordability itself has not been measured. Discount offers should thus be "trialled" and not viewed as sure causes for departures.

Cluster 1: Established Low-Churn Customers

The Churn rate of Cluster 1 is 4.80% and 1,124 customers are in this cluster. What sets it apart from the higher-risk group is that people tend to stay for longer within it and charges are less standardised. A steady maintenance fee and relationship; this is more defensible than a general reasoning of discount retaining. This cluster can also be used as a benchmark to review the difference in the service quality and journey.

Cluster 2: Longer-Tenure, Higher-Charge Customers

In Cluster 2 there are 1,882 customers, 15.67% of which churn and which have higher than average standardised monthly charges. Should not be called up the being the highest risk senior, simply because the older report said that. Appropriate consumer behavior is to guard the service supply, observe billing issues and look into specific incentives just for those having increased estimated risk.

Cluster 3: Shorter-Tenure, Highest-Churn Customers

Cluster 3 has 2,146 customers and the highest churn at 48.84%. This group uniquely tops the list for investigation of retention, in terms of its shorter relative tenure and above average standardised charges. Support with onboarding and initial use reviews and plan-value checks could be piloted. The relationship between the variables observed does not necessarily demonstrate that the monthly charges, or the change of tenure are in and of themselves the reason for cancellation.

8. ANN Predictive Modelling

The full and verified modelling pipeline is used in the predictive phase instead of the earlier two-layer result of the sample document, based on 10 features. All 16 segment model inputs are given to each candidate after having produced 4 train-only segment indicators. Dense layers applied with dropouts and sigmoid probability output is used in the ANN. In the training, the parameters of Adam optimisation and binary cross-entropy and early stopping are used. Completed the reference run, which took 21 epochs of a total of 45 epochs.

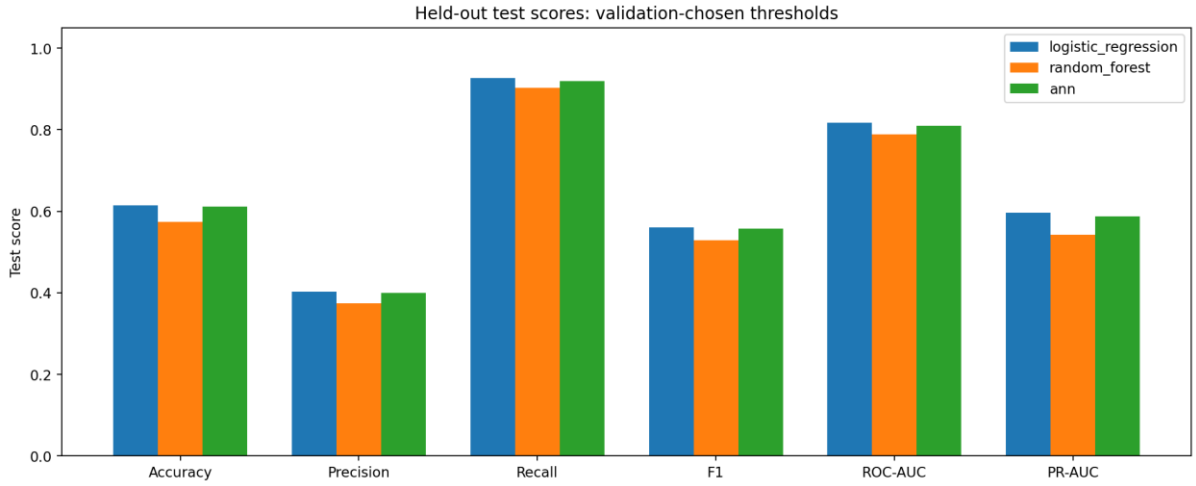
Table 5. Model design and evaluation choices

Design choice	Implemented approach
Predictors	12 encoded source fields + 4 train-only cluster flags
ANN loss / optimiser	Binary cross-entropy / Adam
Stopping rule	Early stopping; 21 epochs completed
Candidate models	ANN, logistic regression, random forest
Selection criterion	Validation average precision
Threshold criterion	Validation F2 maximisation
Test set	1,349 independently held-out observations

Since the accuracy of the ANN was close to logistic regression, being the validation Average Precision of 0.607, it was selected. Although later logistic regression did marginally better on the held out test data, this is a proper decision for this validation-time. The work flow is built on top of the model-fitting and early-stopping facilities, as documented in the Keras Team (n.d.). The following figures were created using a set of tools described by Matplotlib Development Team (n.d.).

Table 6. Held-out performance using each validation-selected threshold

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
ANN	61.23%	39.98%	91.90%	55.72%	0.810	0.587
Logistic regression	61.53%	40.24%	92.74%	56.13%	0.817	0.596
Random forest	57.45%	37.47%	90.22%	52.95%	0.789	0.542

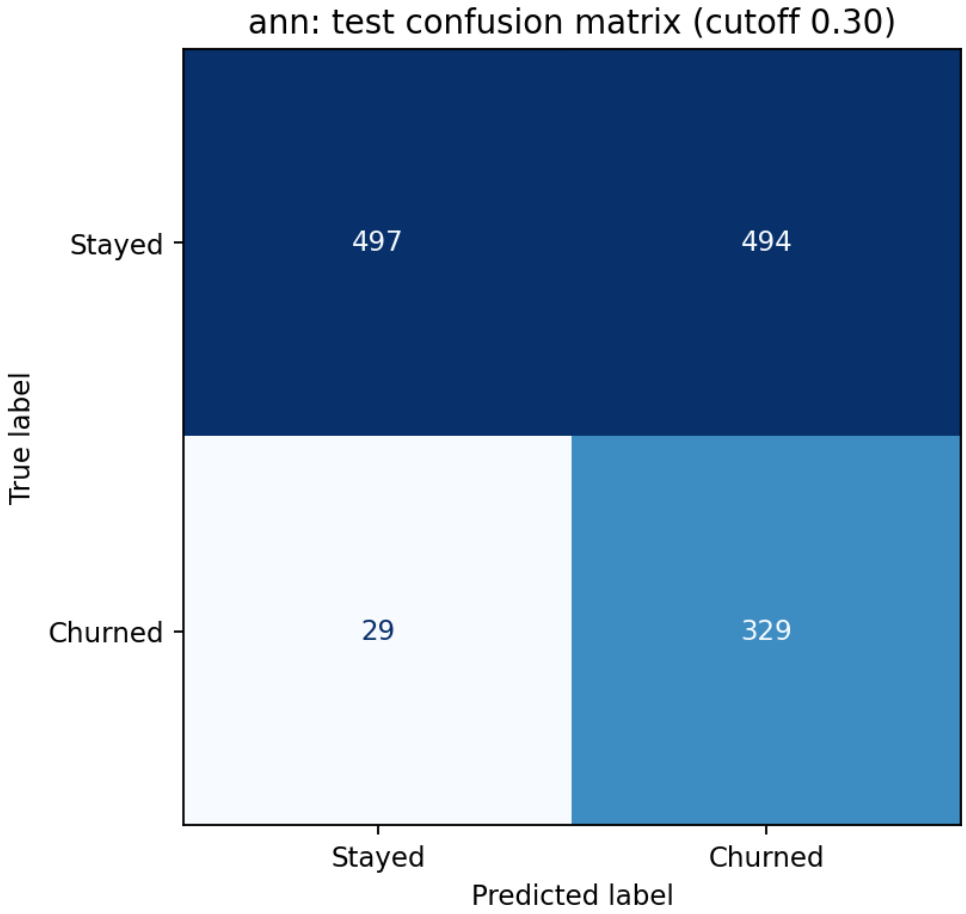


8.1 Confusion Matrix and Business Interpretation

There are 329 true churners identified by the ANN and 29 are not identified by the ANN. It also identifies 494 non-churners as churners, and correctly identifies 497 as non-churners. So with 91.90% recall it does not necessarily mean that 91.90 per cent of alerts are correct. The percentage of confirmed churners (held-out labels) is 39.98% among the alerted customers. A lot of individuals would be called for a "false positive" retention service for this threshold, causing retention service cost and customer experience problems.

Table 7. ANN confusion matrix and operational interpretation

Classification	Count	Operational meaning
True positive	329	Churner reached by risk flag
False negative	29	Churner not flagged
False positive	494	Non-churner flagged
True negative	497	Non-churner correctly not flagged



The same ANN was able to achieve an accuracy of 71.98% and recall of 75.70% at the fixed value of 0.50. On its own, the point is not necessarily a fault in the model, but rather a business decision to run the machine at a higher recall operating point to provide greater recall, sacrificing accuracy. Key considerations for the decision must include intervention costs, as well as available contact capacity and the value of kept customers.

9. Combined Factors Contributing to Churn

Churn is related to segment characteristics, senior status, tenure and transformed charges. There is a shorter relative tenure and higher loss in cluster 3. When it comes to customers' churn rates, 41.24% of seniors churned across the file compared to 23.61% of all customers. These patterns are used for describing and NOT for creating causes.

Table 8. Evidence and practical interpretation of churn indicators

Observed indicator	Evidence	Reasonable next check
Segment membership	Cluster 3 churn 48.84%	Review early customer journeys
Tenure	High-risk cluster has shorter relative tenure	Test onboarding interventions
Monthly charges	High-risk cluster above standardised mean	Review perceived plan value
Senior status	41.24% churn for seniors overall	Assess support accessibility
Predictive score	ANN recall 91.90% at chosen cutoff	Balance alerts against contact cost

Caution must also be raised to contract commitment. The file already encoded doesn't replicate each and every typical relationship generally documented in telecommunications case studies. So there can be no claims that it was lost due to month to month contracts. Before making the transition from correlation to specific explanation(s), stakeholder interviews, complaint histories and controlled offer tests should be undertaken.

10. Retention Strategy Recommendations

Retention spend needs to be prioritized based on a blend of forecast risk, actual segment profile and potential pocketbook size. The first idea that was proposed is to offer controlled early-relationship support by invitation to high scoring customers from published Cluster 3. Company should perform service and value tests for the Cluster 2 only when there are individual risks. Reliable service must be managed to lower-risk clusters without an automatic monetary incentive.

Table 9. Targeted customer retention proposal

Group	Priority	Suggested intervention	Evaluation measure
Cluster 3	Highest	Early engagement, plan-value review and targeted contact	Incremental churn reduction
Cluster 0	Conditional	Affordability message for high-scoring individuals	Retention per offer cost
Cluster 2	Selective	Billing clarity and service review	Complaint reduction and retention
Cluster 1	Maintenance	Service continuity and loyalty recognition	Stable churn and satisfaction
All high-score customers	Operational	Ranked and capacity-limited retention list	Precision of contacts and net value

10.1 Recommended Retention Campaign Design

The campaign strategy should start by stating the eligibility rule (top 20% of modellers to receive the call) and the budget available to make contacts, and the published eligibility segments as well. Customers will be randomly allocated to a proposed intervention and/or to a standard service, with proper ethical and privacy protections in place. It should have the same window of outcomes for both groups. Retained revenue should be calculated in accordance with following those incentives and contract costs, and shouldn't be shown only as the volume of successful conversations.

10.2 Why Cluster 3 Should Be the First Priority

First is Cluster 3, which has the highest number of customers, churning accounted for 1,048 of the customers observed and has almost double the churn rate of the overall baseline. Data in these facts replace the older sample's finding that Cluster 2 was the highest risk group. There must still be a balance between Cluster 3 priority and predicted risk, ability to contact customers and the expected economic benefit. The strategy should be tested prior to any large investment in an outreach campaign.

11. Implementation Roadmap and Stakeholder KPIs

The proposed 90-day roadmap sees predictive modelling as a decision making process, not a turnkey service to be deployed for production. Business owners should check the descriptions of their data, and approve risk, contact and customer-service protections during a first month. A small controlled intervention should be given for the second month. Outcome should be checked and the threshold redone (if required) and the decision made on expansion made in the third month.

Table 10. Implementation roadmap

Period	Main action	Acceptance evidence
Days 1–30	Validate field provenance and agree pilot rules	Approved targeting and monitoring plan
Days 31–60	Conduct controlled Cluster 3 outreach	Complete contact and outcome records
Days 61–90	Review uplift, false alerts and cost	Decision on controlled expansion
Quarterly thereafter	Monitor drift, retrain and review thresholds	Stable performance and measured net benefit

11.1 Suggested KPIs

The main business insights are incremental churn reduction against a matched control. Supporting measures are net revenue after contact and discount costs, segment-level churn, customer complaint rate and positive contact outcomes. Some parameters that should be monitored technically: PR-AUC, churn recall, alert precision, calibration and false positive rate. The following measures must all be monitored in tandem so that if there was an improvement in one there was not an improvement in the whole business.

12. Project Limitations

There are several limitations that affect the conclusions that can be drawn. The dataset does not consist of a full telecom history but is rather a prepared dataset. Its tenure and monthly fees were regularised upstream and similarly there is no original raw to standardised transfomer. Thus, it is not possible to report original currency amounts and pre-split scaling leakage is not eliminated. The results of the modelling can also change depending on how the upstream source is reconstructed.

Published cluster IDs and PCA co-ordinates were calculated on all the data prior to the supervised split. Thus, they are not included in model predictors. A distinctively train-only clustering method enhances the quality of evaluation discipline; however, this method gives different labels to published segments. The project additionally does not have a complaint history, service disruption record, client satisfaction measures and interventions outcomes. Therefore the outlined retention procedures have not shown a real decrease in cancellation.

13. Future Improvements

Future efforts should "rediscover" the pre-processing from raw data, and reprocess all pre-processing from original data records. If appropriate dates are available, then there could be a case for considering time-based validation, which could better reflect future churn. Calibrations and retention-cost simulation together with behavioural data, would be useful in selecting more realistic decision thresholds.

An incremental uplift should then be measured as part of a controlled campaign, with transparent stakeholder reporting. Other forecasting approaches can be investigated if they provide greater performance on an independent, pre-specified future test set. But the current logistic regression standard-of-truth is marginally better than the ANN, with higher held out ROC-AUC and PR-AUC. A more optimal deployment design and/or an assessment of intervention strategies may thus be more valuable than raising network complexity. Technical analysis has been made reproducible using the runtime specified in Google (n.d.).

14. Conclusion

The finished product will present a comprehensive picture of churn from 6,741 completed telecom records. The most churned Customer Cluster (Cluster 3) had 2146 customers according to published customer segmentation, which also is the biggest Customer Cluster. An alternative modelling process bypasses the entire concept of using these clusters, derived by using the same information globally, as inputs for predictions, and compares the performance of an ANN with two baseline methods. The ANN can identify 91.90% of the churners for the chosen threshold value and has many false alarms. Held-out ranking metrics with logistic regression tested is slightly better.

So a sound business response is to use predictions without projecting false confidence. The information contained in the historical segments can be used to create a meaningful retention offer, and model probabilities to prioritise outreach. To determine whether early intervention does actually result in an increase in customer retention, the testing must be controlled and cost conscious. Damage the original transformations of the data, monitoring model drift and checking for results would bring the project together prior to operational deployment.

References

Goreja, A. (n.d.) Stage_2_Deliverables. GitHub repository. Available at: https://github.com/aadeshgoreja/Stage_2_Deliverables (Accessed: 25 October 2026).

Scikit-learn Developers (n.d.) User Guide. Scikit-learn. Available at: https://scikit-learn.org/stable/user_guide.html (Accessed: 25 October 2026).

Keras Team (n.d.) Keras 3 API documentation. Available at: https://keras.io/api/ (Accessed: 25 October 2026).

Pandas Development Team (n.d.) User guide. pandas documentation. Available at: https://pandas.pydata.org/docs/user_guide/index.html (Accessed: 25 October 2026).

Matplotlib Development Team (n.d.) Matplotlib documentation. Available at: https://matplotlib.org/stable/ (Accessed: 25 October 2026).

Google (n.d.) Google Colaboratory documentation. Available at: https://colab.research.google.com/notebooks/intro.ipynb (Accessed: 25 October 2026).

Appendix A: Data Dictionary and Encoding Notes

Table 11. Principal dataset fields and encoding

Field	Representation	Analytical meaning
Churn	Binary 0/1	Observed outcome
gender	Numeric indicator	Customer demographic field
SeniorCitizen	Binary 0/1	Senior status
Dependents	Binary 0/1	Dependent status
tenure	Pre-standardised numeric	Relative relationship duration
MonthlyCharges	Pre-standardised numeric	Relative charge level
PhoneService	Binary indicator	Phone access
MultipleLines	Binary indicator	Multiple line service
InternetService_DSL	Binary indicator	DSL service
InternetService_Fiber optic	Binary indicator	Fibre service
Contract_*	Three indicator columns	One-hot contract attributes
Cluster	Integer 0–3	Published descriptive grouping
PCA1, PCA2	Continuous coordinates	Published visual projection only